



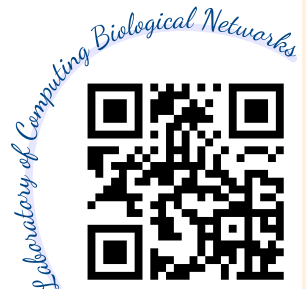
Lecture 15

2025-12-10

Synaptic plasticity; Hebbian learning; Bienenstock-Cooper-Munro rule; Spike-timing plasticity

- Online slides: lec15-synaptic_plasticity.html
- Code: code15.ipynb
- Homework: hw15.pdf

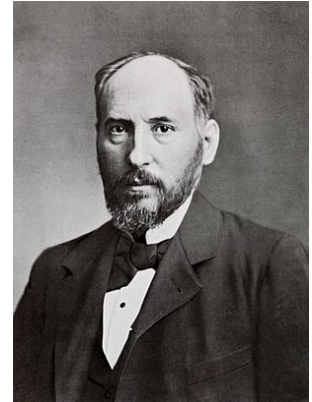
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How activity affects synapses

Synaptic plasticity

- Believed to underlie learning and memory
- Crucial for development of neural circuits
- Example tasks: auto-/hetero-associative memory, pattern recognition, storage and recall, temporal sequences, function approximation



Hebbian learning

“When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, ... A’s efficiency, as one of the cells firing B, is increased.”

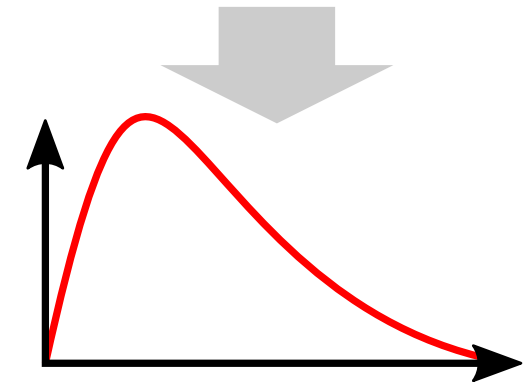
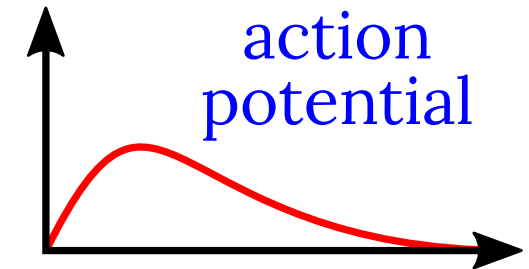
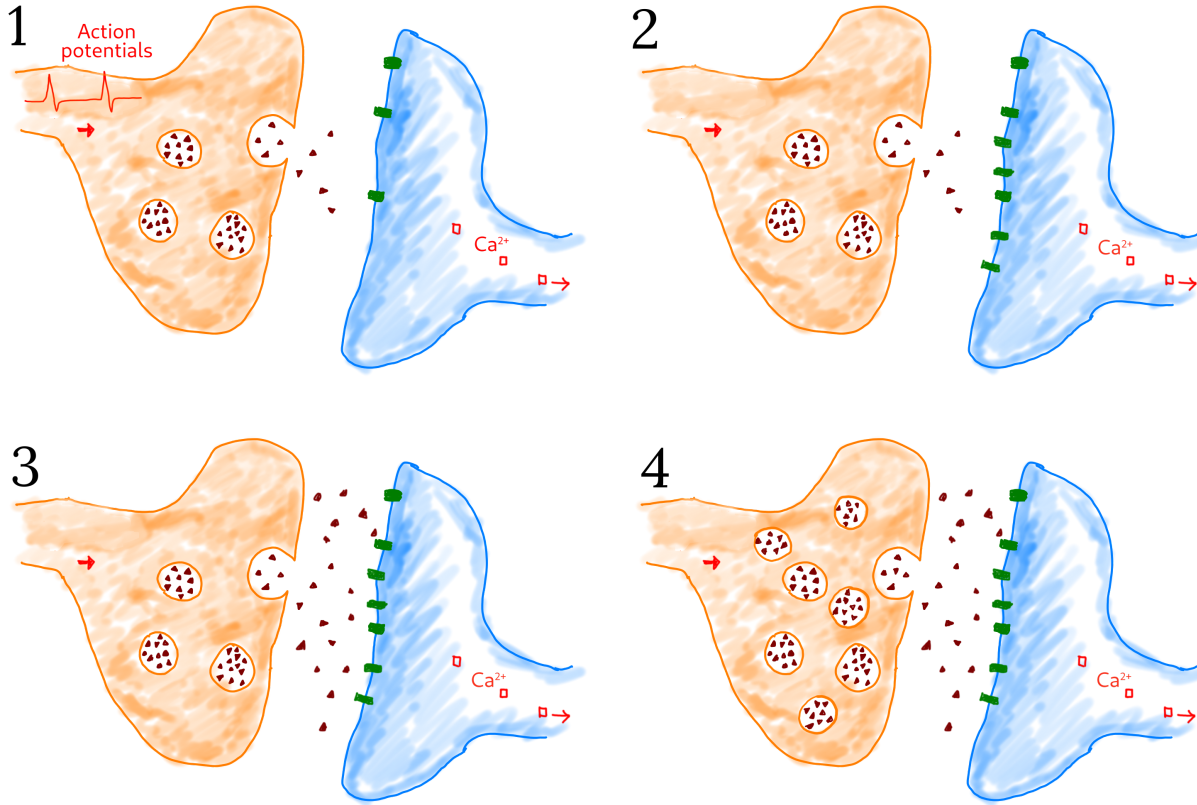
Hebb (1949)



Simplified version: *“Cells that fire together, wire together.”*

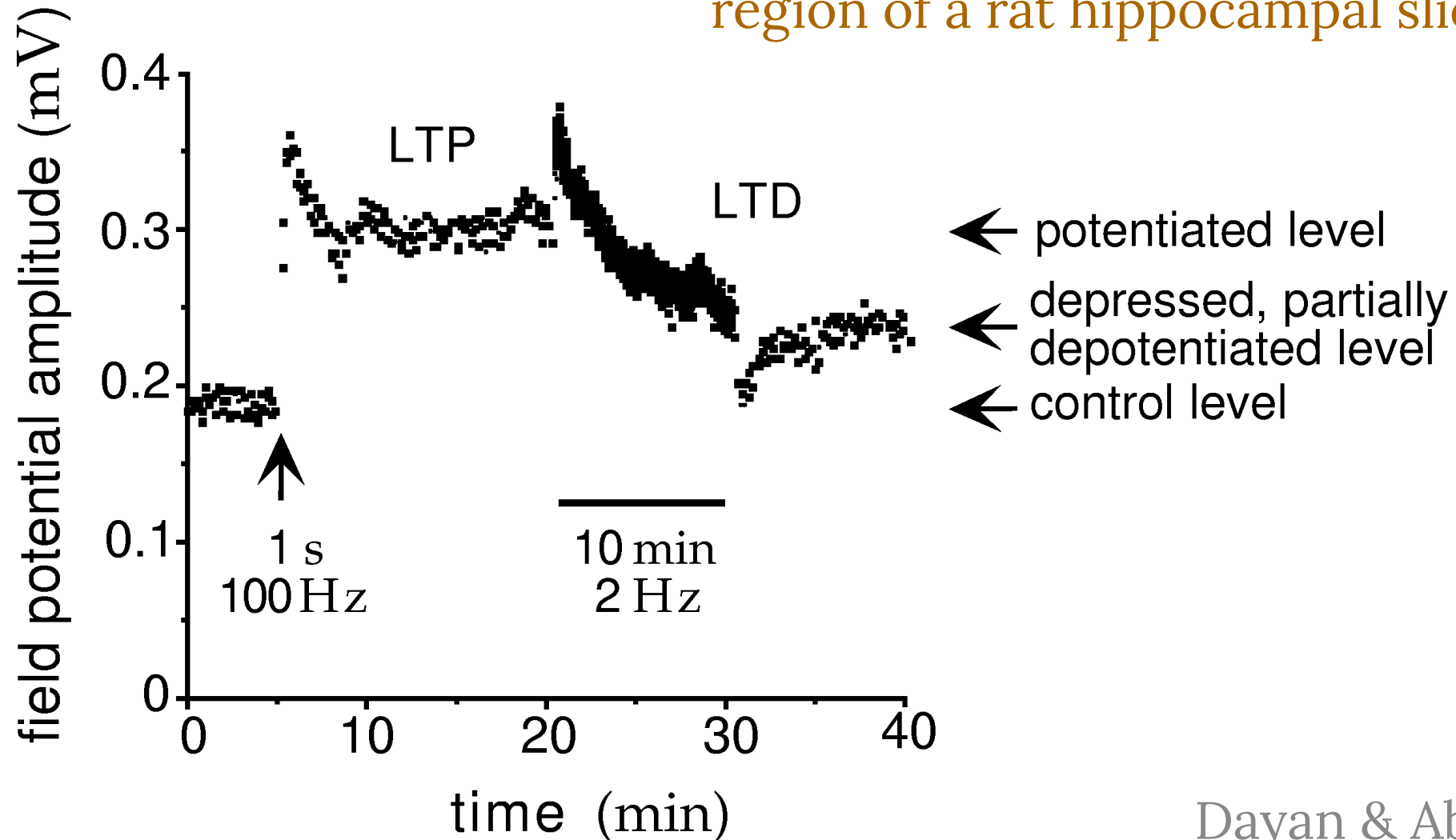
Long-term potentiation (LTP)

1. Synapse is repeatedly stimulated
2. More dendritic receptors
3. More neurotransmitters
4. Stronger link between neurons



LTP and LTD at CA1

Schaffer collateral inputs to CA1
region of a rat hippocampal slice



Dayan & Abbott

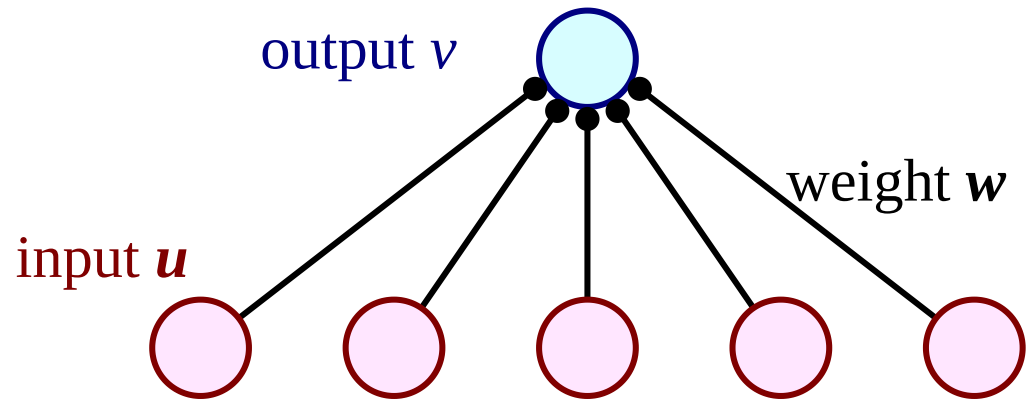
Synaptic plasticity rules

Firing rate model

$$\tau_r \frac{dv}{dt} = -v + \mathbf{w} \cdot \mathbf{u}$$

Row vector

$$= -v + \sum_{b=1}^{N_u} w_b u_b$$



Steady state:

$$v = \mathbf{w} \cdot \mathbf{u}$$

Synaptic modification is included by specifying how w changes as a function of the pre- and postsynaptic activity.

The basic Hebb rule

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u}$$



$$\tau_w \frac{d\mathbf{w}}{dt} = \langle v\mathbf{u} \rangle$$

Averaged Hebb rule

$\langle \dots \rangle$: Average over ensemble of input patterns

For steady state $v \rightarrow \mathbf{w} \cdot \mathbf{u}$,

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w} \quad \text{or} \quad \tau_w \frac{dw_b}{dt} = \sum_{b'=1}^{N_u} Q_{bb'} w_{b'}$$

where $\mathbf{Q} = \mathbf{u}\mathbf{u}$ or $Q_{ab} = \langle u_a u_b \rangle$ is the **input correlation matrix**.

Outer product

Discrete process for plasticity

$$\tau_w \frac{\mathbf{w}(t + \Delta t) - \mathbf{w}(t)}{\Delta t} = \mathbf{u}v \Rightarrow \mathbf{w}(t + \Delta t) = \mathbf{w}(t) + \frac{\Delta t}{\tau_w} \mathbf{u}v$$

Step-by-step update: $\mathbf{w}_{i+1} = \mathbf{w}_i + \epsilon \mathbf{u}v$

$$\mathbf{w} \rightarrow \mathbf{w} + \epsilon \mathbf{Q} \cdot \mathbf{w}$$

 Learning rate

Effect of Hebb plasticity (correlation rule)

$$\frac{d|\mathbf{w}|^2}{dt} = 2\mathbf{w} \cdot \frac{d\mathbf{w}}{dt} = \frac{2}{\tau_w} \mathbf{w} \cdot \mathbf{u}v = \frac{2v^2}{\tau_w} \geq 0$$

Always grows! Need additional constraints or saturation rules

Adding LTD (covariance rule)

Thresholds for potentiation

$$\tau_w \frac{d\mathbf{w}}{dt} = (v - \theta_v) \mathbf{u} \quad \text{or} \quad \tau_w \frac{d\mathbf{w}}{dt} = v(\mathbf{u} - \theta_u)$$

Using average as the threshold

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{u}(v - \langle v \rangle)$$

Consider average effect

$$\tau_w \frac{d\mathbf{w}}{dt} = \langle \mathbf{u}(v - \langle v \rangle) \rangle = \mathbf{C} \cdot \mathbf{w}$$

where $\mathbf{C} = \langle (\mathbf{u} - \langle \mathbf{u} \rangle)(\mathbf{u} - \langle \mathbf{u} \rangle) \rangle$ is the covariance matrix.

The Bienenstock-Cooper-Munro (BCM) rule

Long-time effect of covariance rule $\frac{d|\mathbf{w}|^2}{dt} = 2v(v - \langle v \rangle) \Rightarrow \langle v^2 \rangle - \langle v \rangle^2 \geq 0$

Still unstable

BCM rule

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u}(v - \theta_v)$$

with sliding threshold

$$\tau_\theta \frac{d\theta_v}{dt} = v^2 - \theta_v$$

changes slower than input pattern but faster than weights

Cascade of time scales: $\tau_r < \tau_\theta < \tau_w$

☆ Competition between synapses is implemented

Regulation of plasticity

- Fixing sum or **sum of squares of weights** (rigid or dynamic)
- **Subtractive** normalization: (**$\mathbf{n} \cdot \mathbf{w}$ is conserved**)

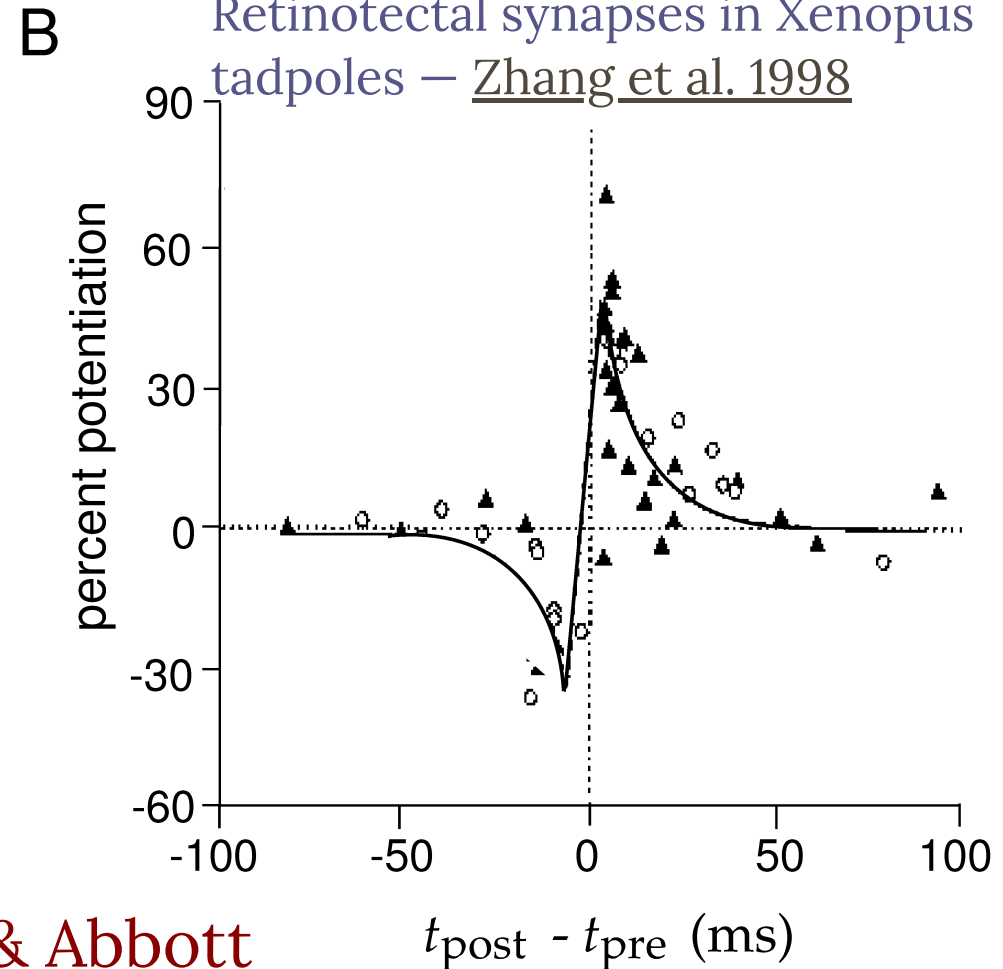
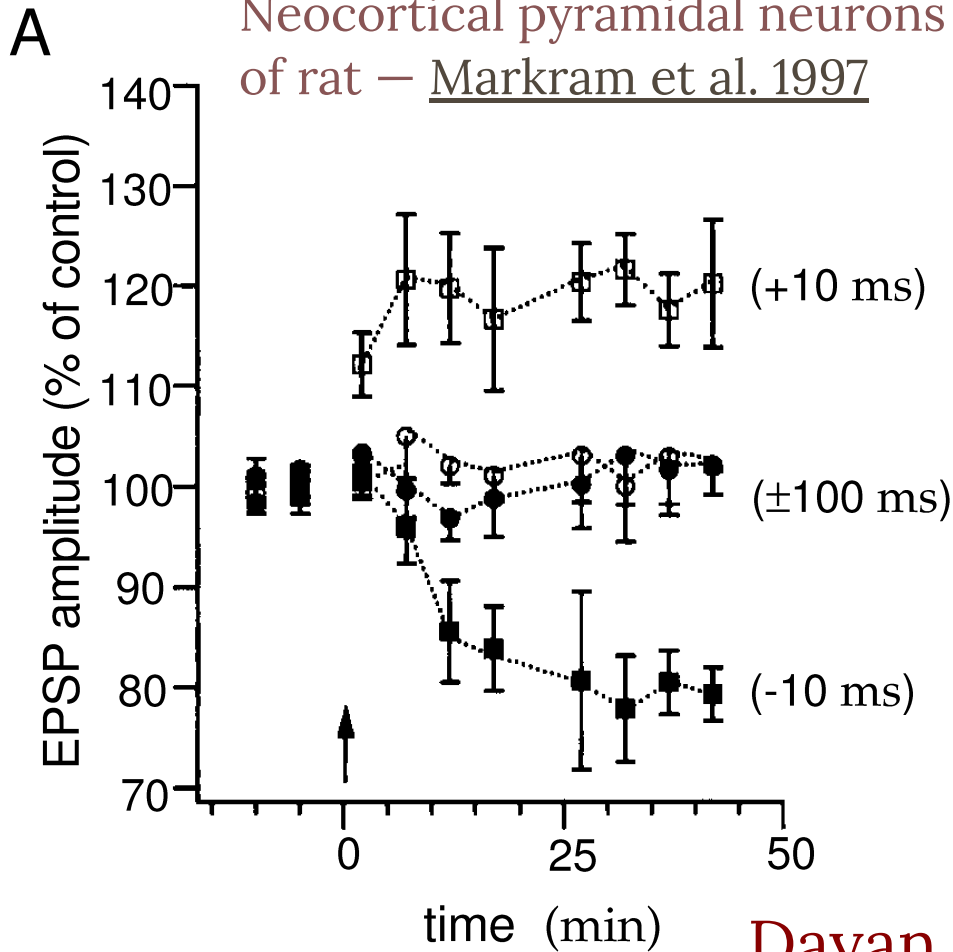
$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u} - \frac{v(\mathbf{n} \cdot \mathbf{u})\mathbf{n}}{N_u}$$

- **Multiplicative** normalization and the Oja rule

$$\tau_w \frac{d\mathbf{w}}{dt} = v\mathbf{u} - \alpha v^2 \mathbf{w}$$

$$\begin{aligned} \text{(applying } \cdot \mathbf{w}) \Rightarrow \tau_w \frac{d|\mathbf{w}|^2}{dt} &= 2v^2 (1 - \alpha|\mathbf{w}|^2) \\ &\Rightarrow |\mathbf{w}|^2 \text{ relaxes to } 1/\alpha \end{aligned}$$

Timing-based rules



Dayan & Abbott

LTP and LTD produced by 50–75 pairs of pre- and postsynaptic action potential with various timings.

Unsupervised learning in bio-systems

Major areas of concerns

- Development of neuronal selectivity
- Formation of cortical maps

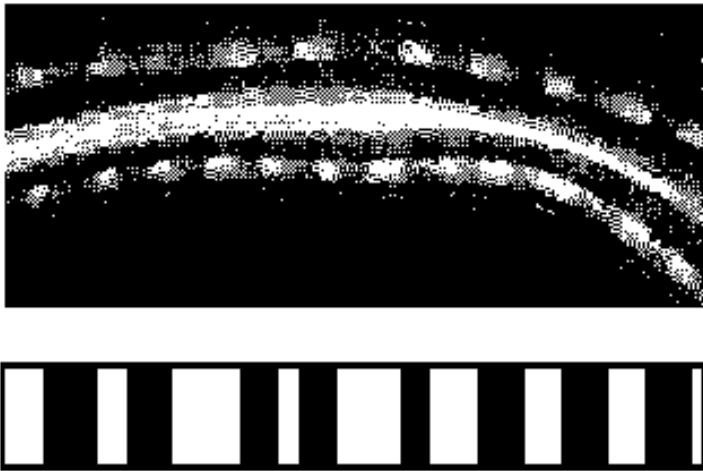
Stages of neural development

- Initial targeting of axons: layer for innervation, coarse patterning of projection — activity independent
- Refining, local organization — can be activity dependent

Activity-dependent plasticity models

Ocular dominance

- Input to neurons favors one eye over the other
- Example: LGN input to layer 4 neurons
- Alternates across cortex forming stripes



Cat primary visual cortex
radioactively labeled to distinguish
regions activated by one eye or the
other.

How to solve plasticity equations

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w}$$

Resort to eigenvectors & eigenvalues. Expand $\mathbf{w}$ in eigenvectors:

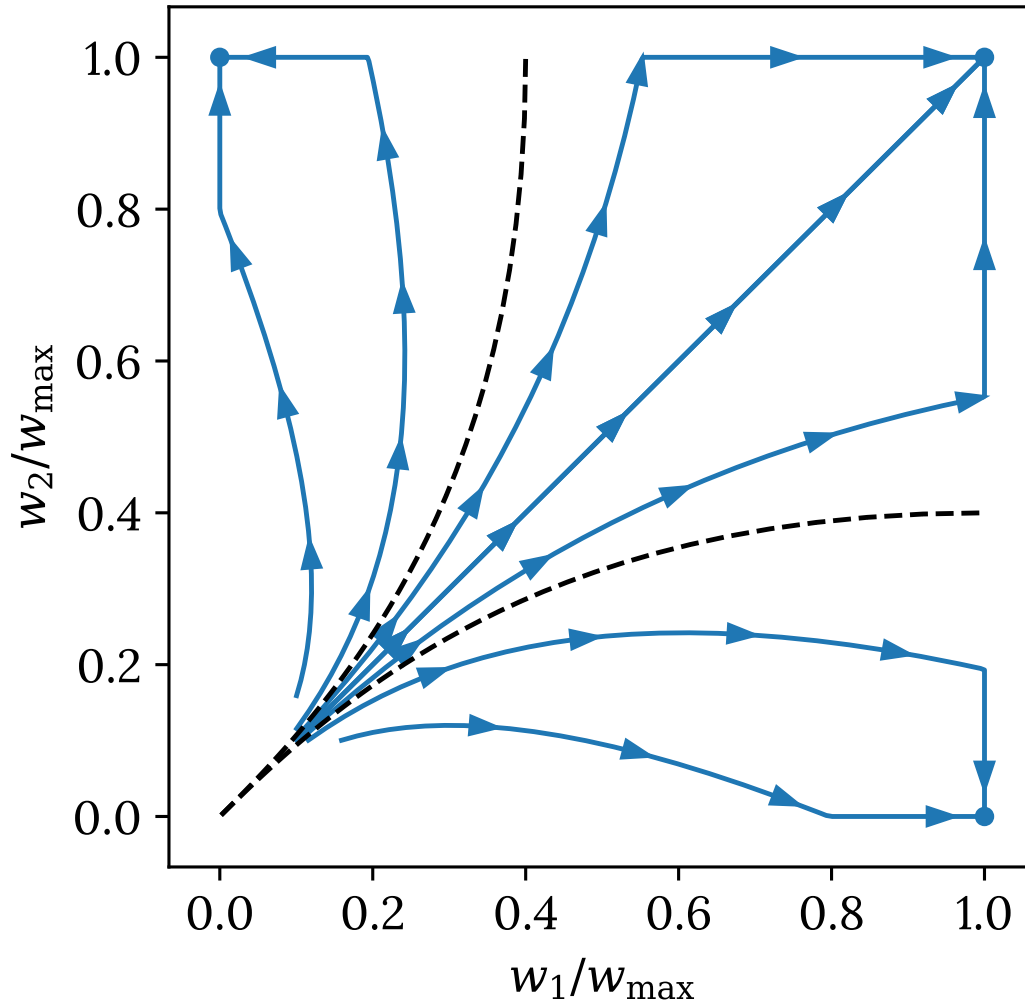
$$\mathbf{w}(t) = \sum_{\mu=1}^{N_u} c_{\mu}(t) \mathbf{e}_{\mu}$$

Solution:

$$\mathbf{w}(t) = \sum_{\mu=1}^{N_u} \exp \frac{\lambda_{\mu} t}{\tau_w} (\mathbf{w}(0) \cdot \mathbf{e}_{\mu}) \mathbf{e}_{\mu}$$

Principal eigenvector: $\mathbf{e}_1$ After training, $v \propto \mathbf{e}_1 \cdot \mathbf{u}$

Problem with saturation constraint



Input correlation matrix:

$$\mathbf{Q} = \begin{pmatrix} 1 & -0.4 \\ -0.4 & 1 \end{pmatrix}$$

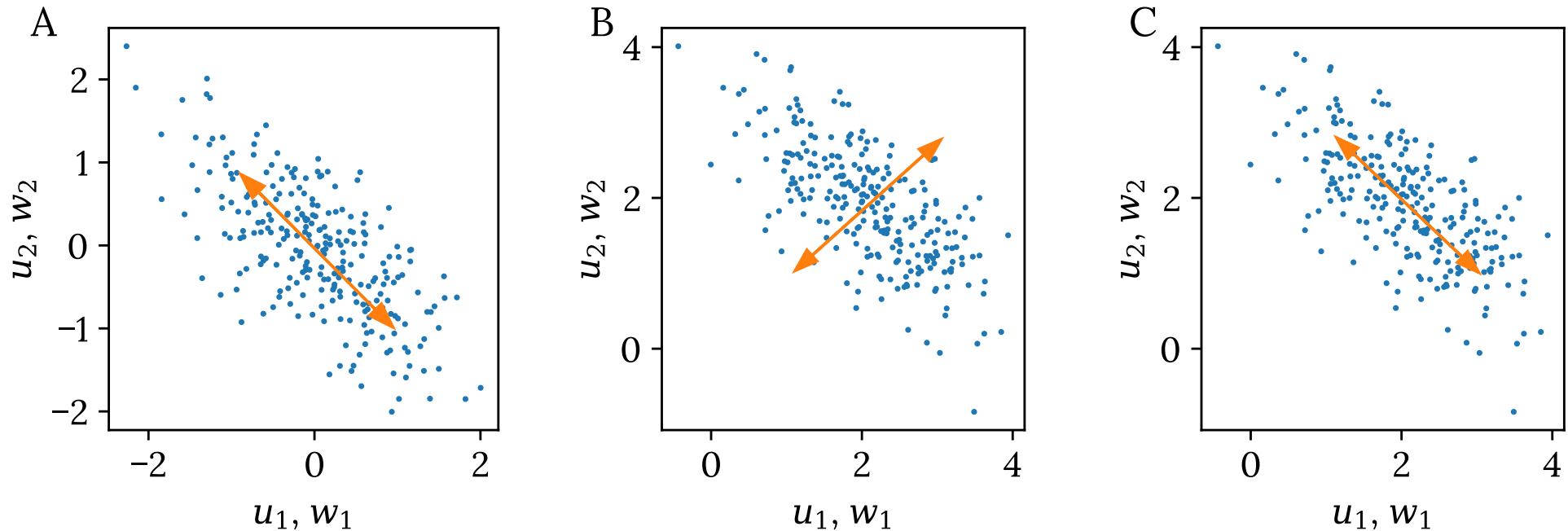
Principal eigenvector

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

fix: Use the Oja rule

$$\mathbf{w}(t \rightarrow \infty) = \frac{\mathbf{e}_1}{\sqrt{\alpha}}$$

Principal component projection



A. Input $\mathbf{u} = (u_1, u_2)$; Stationary synaptic weight $\mathbf{w} = (w_1, w_2)$

B. Correlation-based modification

C. Covariance-based modification

Hebbian development of ocular dominance

$$v = w_R u_R + w_L u_L$$

$$\mathbf{Q} = \langle \mathbf{u}\mathbf{u} \rangle = \begin{pmatrix} q_S & q_D \\ q_D & q_S \end{pmatrix} \quad \text{with} \quad \begin{aligned} q_D &= \langle u_R u_L \rangle \\ q_S &= \langle u_R u_R \rangle = \langle u_L u_L \rangle \end{aligned}$$

Diagonalization:

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{e}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

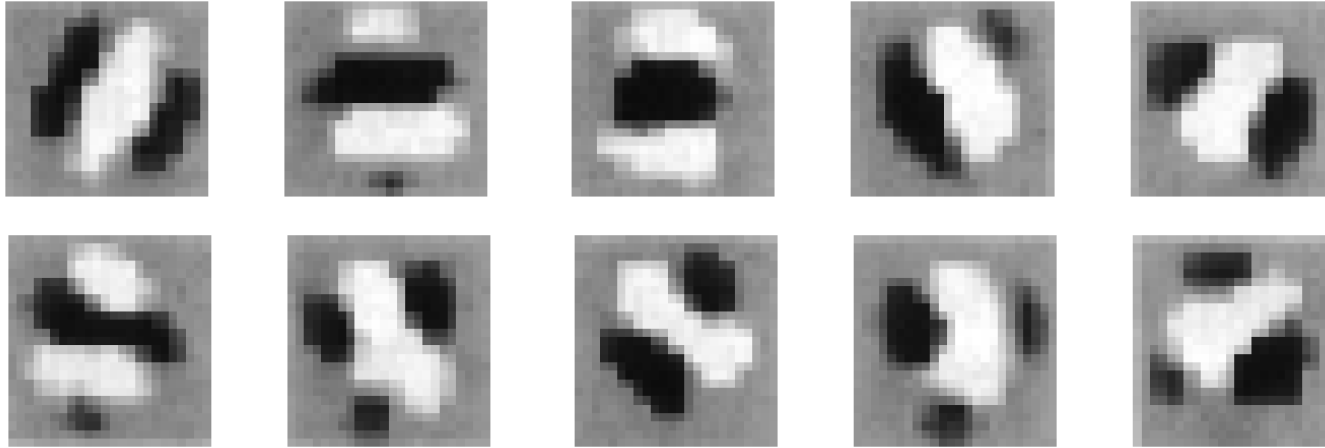
Eigenvalues: $\lambda_1 = q_S + q_D$, $\lambda_2 = q_S - q_D$

λ_1 dominates $\rightarrow$ no ocular dominance

Applying subtractive normalization favors λ_2 and results in dominance.

$$\begin{aligned} w_+ &= w_R + w_L \\ w_- &= w_R - w_L \\ &\Rightarrow \text{decouple} \end{aligned}$$

Development of orientational selectivity



Miller 1994

Essential ingredients

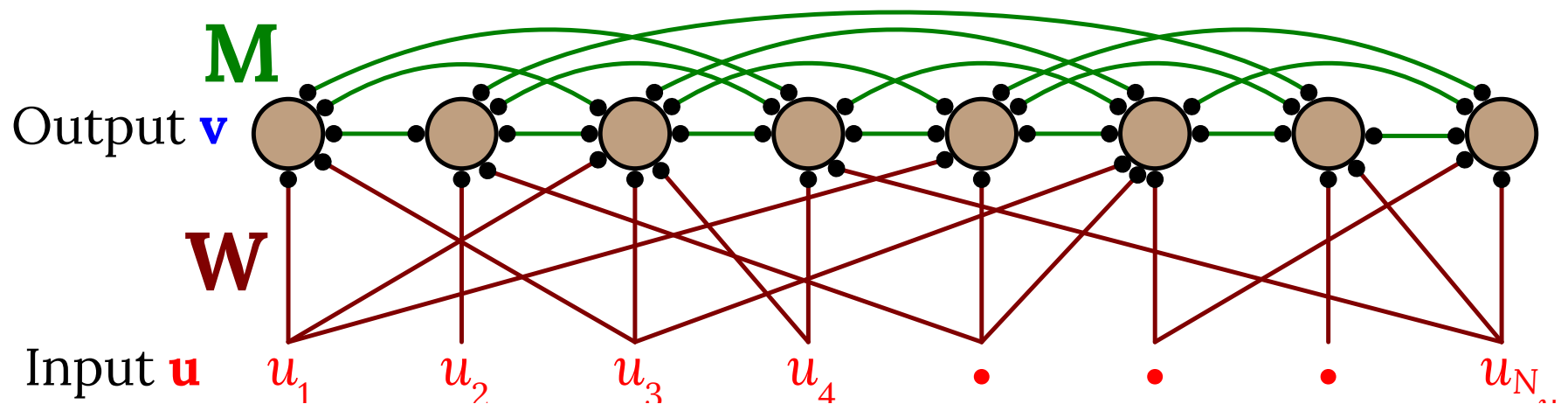
- Large array of ON- and OFF-center LGN afferents
- Correlation-based development model
- Growing non-uniform and non-isotropic eigenvectors
- Proper constraints such as localization through arbor function
- Nonlinearity!

Multiple post-synaptic neurons

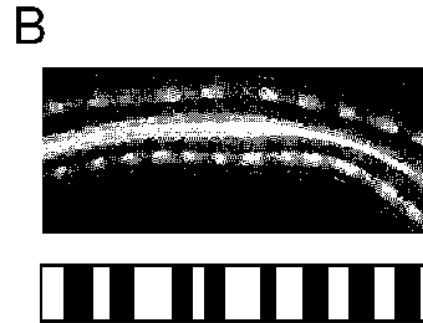
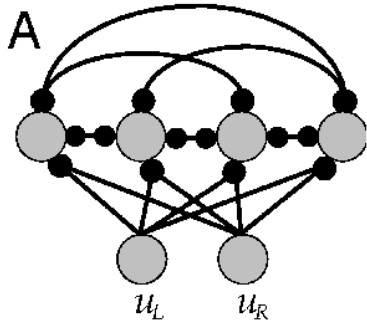
$$\tau_r \frac{d\mathbf{v}}{dt} = -\mathbf{v} + \mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}$$

Steady state: $\mathbf{v} = \mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}$

Solution: $\mathbf{v} = \mathbf{K} \cdot \mathbf{W} \cdot \mathbf{u}$ with the inverse $\mathbf{K} = (\mathbf{I} - \mathbf{M})^{-1}$



Development of ocular dominance stripes



$$\tau_w \frac{d\mathbf{W}}{dt} = \langle \mathbf{v} \mathbf{u} \rangle = \mathbf{K} \cdot \mathbf{W} \cdot \mathbf{Q}$$

$$\mathbf{v} = \mathbf{w}_R u_R + \mathbf{w}_L u_L + \mathbf{M} \cdot \mathbf{v}$$

$$\mathbf{w}_+ = \mathbf{w}_R + \mathbf{w}_L, \quad \mathbf{w}_- = \mathbf{w}_R - \mathbf{w}_L$$

$$\tau_w \frac{d\mathbf{w}_+}{dt} = (q_S + q_D) \mathbf{K} \cdot \mathbf{w}_+$$

$$\tau_w \frac{d\mathbf{w}_-}{dt} = (q_S - q_D) \mathbf{K} \cdot \mathbf{w}_-$$

Using subtractive normalization, $\mathbf{w}_+$ is fixed and only $\mathbf{w}_-$ is relevant.

Assumptions:

1. Periodic boundary condition
2. Translation invariance

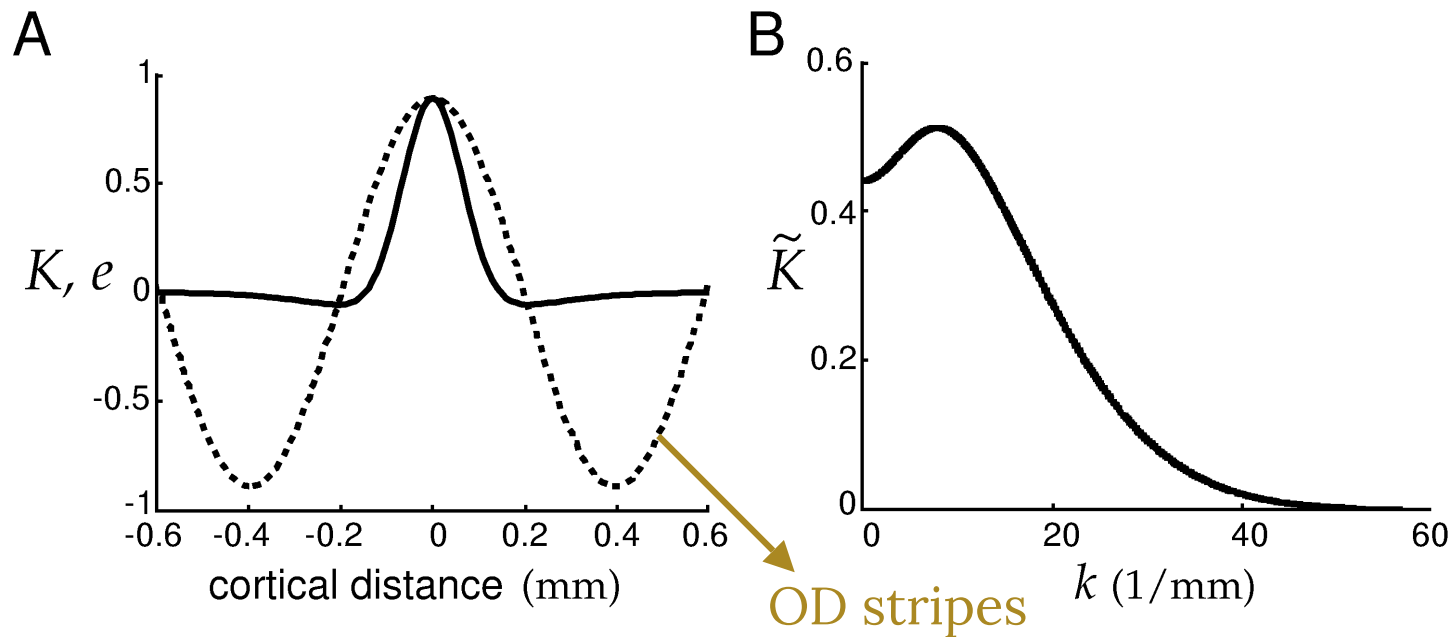
$$\Rightarrow K_{aa'} = K(|a - a'|)$$

Fourier space

With periodic boundary condition, eigenvectors are:

$$e_a^\mu = \cos \left(\frac{2\pi\mu a}{N_v} - \phi \right)$$

$$\mu = 0, 1, 2, \dots, N_v/2$$



Competitive Hebbian learning

Input competition

$$z_a = \frac{(\sum_b W_{ab} u_b)^\delta}{\sum_{a'} (\sum_b W_{a'b} u_b)^\delta}$$

Local cooperation

$$v_a = \sum_{a'} M_{aa'} z_{a'}$$

M is purely excitatory and short range.

